MSc Data Science
DS601: Mathematical Foundations for Data Science
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Tutorial 3

Q1) In an online examination, students can qualify through two attempts.

- 60% of students clear Attempt 1.
- 40% of students fail Attempt 1.
- Among the students who clear Attempt 1, 90% are finally selected.
- Among the students who fail Attempt 1, 70% appear for Attempt 2.
- Of the students who appear for Attempt 2, 80% are finally selected.

A student is randomly selected, and it is found that the student was finally selected.

Question-
What is the probability that the student had failed Attempt 1?

Q2) There are 10 closed doors. Behind one door is a car , while behind the other 9 doors are goats .

You randomly choose Door 1.

The host knows where the car is. He then opens 8 other doors, and all 8 doors contain goats.

Now only two doors remain closed:

- Door 1 — your original choice
- Door 7 — the only other closed door

The host asks:

"Do you want to stay with Door 1 or switch to Door 7?"

(i) What is the probability of winning if you switch to Door 7?

(ii) Which strategy should you choose: Stay or Switch?

(iii)If staying has probability 1/10 of winning the next game become larger because i have already played 200 games? (Gamblers fallacy) (CLASS ACTIVITY)

Q3) A rare disease affects 1% of the population. A diagnostic test is used to detect the disease. The test correctly identifies a person who has the disease as positive with a probability of 98%.

However, among healthy individuals, the test also produces a false-positive result with a probability of 3%.

A randomly selected person undergoes the test and receives a positive result. The doctor recommends a second, independent test. The second test has the same sensitivity and false-positive rate as the first test, and it also returns positive.

What is the probability that the person actually has the disease, given that both tests are positive.

Q4) A manufacturing company operates three production plants P1,P2, and P3, which contribute 30%, 45%, and 25% respectively to the company's total production.

The probabilities that an item produced by these plants is defective are:

$P(D|P1)=0.02, P(D|P2)=0.03, P(D|P3)=0.05$

Each item undergoes an automated quality inspection. The inspection system has the following characteristics:

If an item is defective, the probability that the system identifies it as defective is 95%.

If an item is non-defective, the probability that the system incorrectly identifies it as defective is 4%.

Now, an item is selected randomly from the company's total production and the inspection system identifies it as defective.

Answer the following:

(a) Given that the item is actually defective, what is the probability that it was produced by Plant P3?

(b) Given that the inspection system identifies the item as defective, what is the probability that the item was produced by Plant P3?

(c) If two items are independently selected and both are identified as defective by the inspection system, what is the probability that at least one of them is actually defective?

Q5) 10 doors, but the host gives you a choice:

You choose Door 1.

The host knows where the car is and opens 6 goat doors.

Now 3 doors remain closed:

Your original Door 1

Door 7 Door 10

The host says:

You can stay with Door 1, or switch to Door 7, or switch to Door 10.

Question:

(a)What is the probability that Door 1 has the car?

(b)What is the probability that Door 7 or Door 10 collectively has the car?

(c)If you randomly choose between Door 7 and Door 10, what is your winning probability?

(d)Is switching still better than staying?

Q6) Treatment A recovers 90% of Mild patients and 50% of Severe patients; Treatment B recovers 80% of Mild patients and 30% of Severe patients. For Treatment A, alpha is the Mild-patient proportion. For the numerical Treatment B aggregate, the patient mix used is 80% Mild and 20% Severe.

Using the Law of Total Probability:

(a) Express the aggregated recovery rate of Treatment A, $P(R \mid T = A)$, as a linear formula in terms

of α.

(b) Calculate the exact numerical aggregate recovery rate of Treatment B, $P(R \mid T = B)$.

- Observe that Treatment A is strictly superior to Treatment B within every individual severity subgroup:

$P(R \mid T = A, Z = M) > P(R \mid T = B, Z = M)$ and $P(R \mid T = A, Z = S) > P(R \mid T = B, Z = S)$

Determine the exact interval of values for the allocation parameter α for which Simpson's Paradox

occurs—defined as the reversal where Treatment A appears strictly inferior to Treatment B in

aggregate: $P(R \mid T = A) < P(R \mid T = B)$.

- Suppose researchers conduct a Randomized

Controlled Trial (RCT) such that treatment assignment T is statistically independent of baseline

severity Z:

$P(Z = M \mid T = A) = P(Z = M \mid T = B) = p_M$

Prove algebraically that for any valid population proportion $p_M \in (0, 1)$, the aggregate recovery rate

of Treatment A is strictly greater than Treatment B, demonstrating that randomization prevents

aggregation paradoxes.